

Course Content: M.P.T (MS)-104

Part I

MCPFD 1.1. Clinical Decision Making - Planning Effective Treatment. Clinical decision making models, Team approach, Foundation for clinical decision making.

MCPFD 1.2. Vital Signs. Identification of reasons for monitoring vital signs; importance of monitoring vital signs; common techniques of monitoring vital signs; identification and analysis of normal values with that of abnormal values.

MCPFD 1.3. Principles and application of investigative and imaging techniques in Physiotherapy

- a. Blood test
- b. Arterial Blood Gas (ABG) analysis
- c. Pulmonary Function Test (PFT)
- d. Radiological examination
- e. Computerized Tomography (CT)
- f. Magnetic Resonance Imaging (MRI)
- g. Ultrasonography (US)
- h. Electrocardiography (ECG)
- i. Dope testing

MCPFD 1.4. Evaluation assessment and treatment planning strategies for musculoskeletal, neurological, cardiopulmonary, sports specific and other physiotherapy conditions: Principles of evaluation, clinical manifestations, general and specific clinical examination.

i. Physiotherapy assessment of the following:

- a. Range of motion (ROM)
- b. Tone
- c. Muscular strength and endurance
- d. Flexibility
- e. Coordination - Non equilibrium test - Equilibrium test
- f. Sports specific skills
- g. Cardiac efficiency
- h. Sensory evaluation
- i. Functional Evaluation - Various scoring methods in functional assessment - Validity and reliability
- j. Fitness evaluation - Aerobic - Anaerobic

k. Spasm

l. Trigger Point

m. Tender Point

n. Spasm

ii. Assessment of cognitive, perceptual dysfunctions and vestibular dysfunction.

MCPFD 1.5. Electro-Diagnosis:

- i. Characteristics and components of Electro therapeutic stimulation systems and Electro physiological assessment devices.
- ii. Instrumentation for neuromuscular electrical stimulation.
- iii. Electrical properties of muscle and nerve.
- iv. Neurobiology of afferent pain transmission and central nervous system mechanisms of pain modulation.
- v. Electrical stimulation and circulation.

MCPFD 1.6. Clinical Electro physiological testing: Instruments, Techniques and Interpretations of

- a. Nerve conduction velocity including Repetitive Nerve Stimulation (RNS)
- b. Electromyography
- c. Bio-feedback technique.
- d. Late responses

MCPFD 1.7. Concepts of electro physiological studies in neuro muscular diseases as a diagnostic and therapeutic tool.

MCPFD 1.8. Evoked potentials – VEP, SSEP, MEP, BAEP

Part II

MCPFD 2.1. Psychological aspects of rehabilitation in disability: Psychological tests.

MCPFD 2.2. Developmental Screening

- i. Factors Motor control assessment
- ii. Motor control theories/mechanism
- iii. Patterns of normal development
- iv. specific procedures and tests used to assess motor control defects

MCPFD 2.3. Anthropometry

1. Body measurements - Height - Weight - Circumference
2. Body Proportion - Body Mass Index (BMI) - Waist Hip Ratio (WHR)

3. Body Composition

- i. Somatotyping
- ii. Methods of measurement
 - a. Water displacement method for body fat.
 - b. Skin fold measurement
 - c. Under water weighing
 - d. Bioelectric Impedance Analysis (BIA)

MCPFD 2.4. Differential diagnosis in Physiotherapy

MCPFD 2.5.

- i. Functional evaluation.
 - a. The concepts of health status impairment; functional limitations;
 - b. Disability and Persons with Disabilities;
 - c. Definition of functional activity and the purposes and components of the functional assessment;
 - d. Selection of activity and roles for an individual based on his or her capabilities and functional limitations.
- ii. Various forms of functional tests;
 - a. Physical function test
 - b. Multi-dimensional functional assessment instrument,
 - c. Identification of instrument for testing function.
- iii. Various scoring methods used in functional assessment;
- iv. Reliability and validity of various functional assessments.

MCPFD 2.6. Evaluation of aging